An Overlay for Social Networks based on Directed Acyclic Graphs on the Edge

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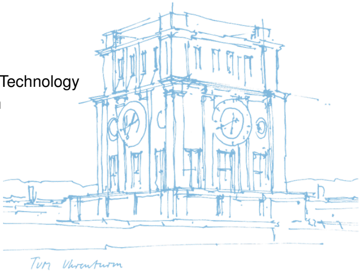
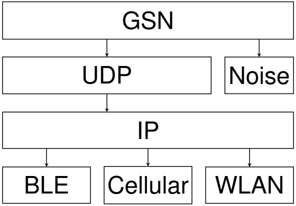
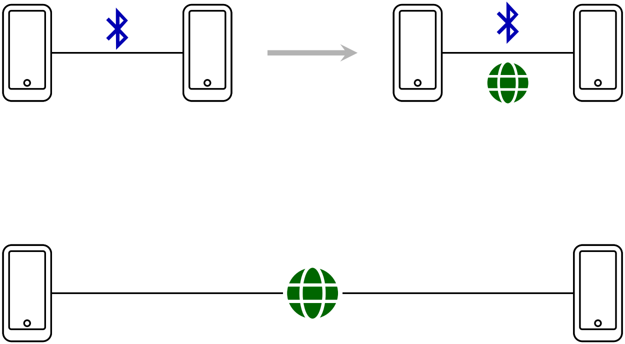


Architecture



- Connect two peer devices according to priority and proximity pair
- Dual-stack network comms: BLE first, Cellular (5G/LTE) and WLAN second
- In close range, use BLE for signaling to establish UDP connection
- ICE-similar stack for address reflection, hole-punching using friends

Dual-stack comms



- BLE as signaling channel
- Advertise multiple addresses
- UDP for hole-punching
- UDX for long-range communication

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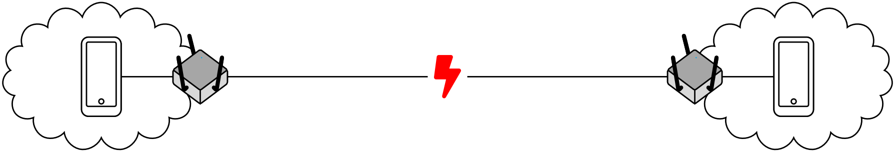
NAT



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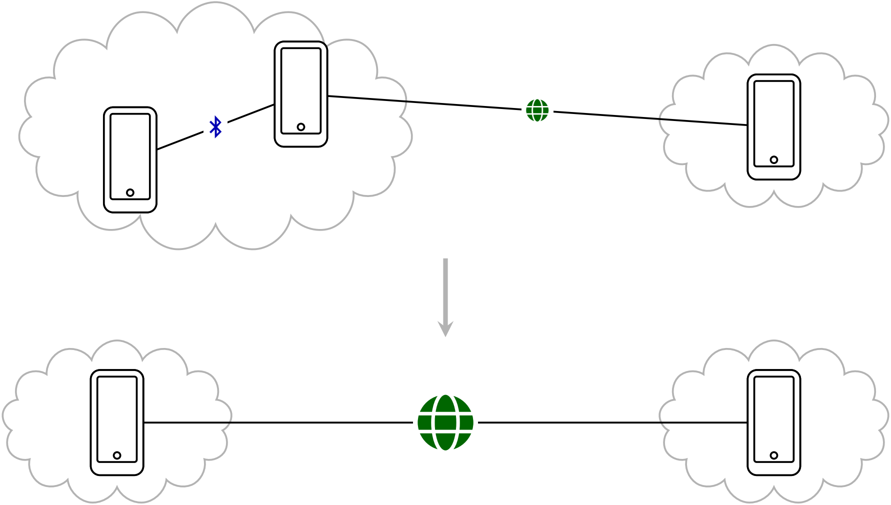
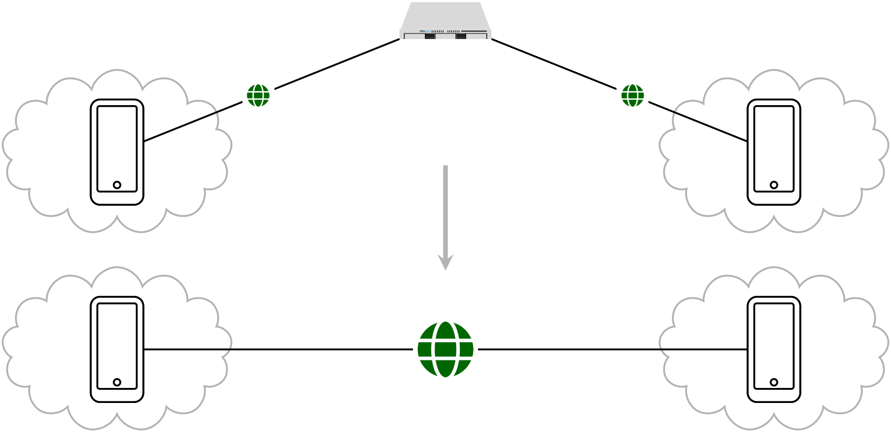
Well-connected friends



Rendezvous Server



Since last weekly

- Migrated BLE communincation layer from two separate central/peripheral plugins to one `grassroots_bluetooth_layer`: unified event model, shared discovery UUID, post-connect ANNOUNCE identity, and BLE role toggle.
- Reworked UDX into a dual-stack Internet transport: IPv4/IPv6 sockets, multi-candidate addresses, live network/public-address refresh, and best candidate-pair selection.
- Aligned rendezvous/NAT traversal with GLP Networking API doc: two-sided punch, RV fan-out (A and B try each others' RV servers when the connection breaks), friends-of-friends mediation (friends can facilitate rendezvous for mutual friends).
- Persist friend data in encrypted local storage so that connections can be initiated post restart.
- Added Noise XX handshake for secure channel encryption: messaging and signaling with Noise XX across clients and rendezvous anchors.
- Hardened delivery over UDX/BLE: ACKs after fragment reassembly, increase max UDX payload size for photos, and stale-session/socket recovery (app loses connections when backgrounded)

Mobility model: city-scale BLE relay

Protocol question

- Given real city encounters, when does sealed BLE relay deliver?
- Compare direct, friend-only, and open relay policies.
- Existing mobility traces give contacts; we measure protocol outcomes.

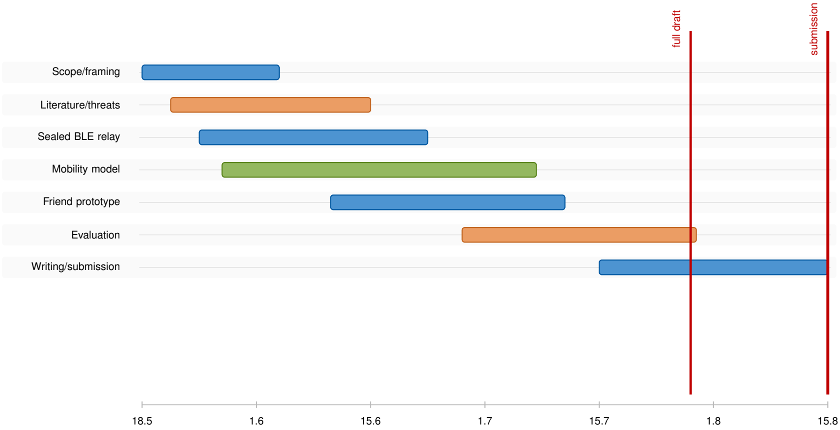
What we track

- Encounters: pseudonym pair, start/end, duration, inter-contact, RSSI band.
- Delivery: created, relayed, ACKed, hop count, TTL expiry, size, failure.
- Cost/risk: relay bytes, duplicates, queues, scan/advertise time, battery drain.

How

- Use existing BLE/contact models as baselines or priors.
- Calibrate with encrypted local trace logs from our Flutter stack.
- Replay traces in simulator with relay policy, TTL, ACKs, and queues.

City-scale BLE thesis



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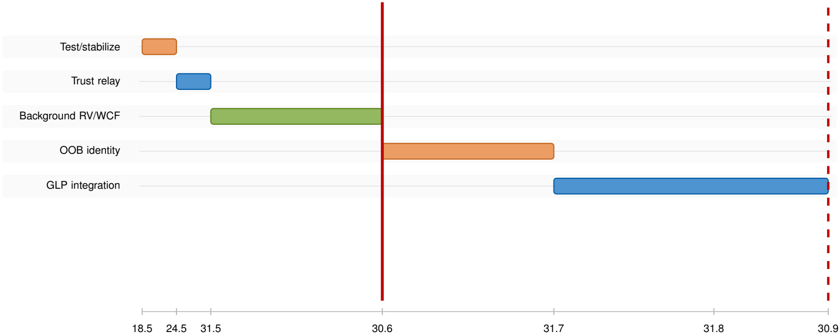
Internet-scale UDX implementation



- **Test/stabilize:** run multiple devices across Wi-Fi, cellular, IPv4/IPv6, NATs, roaming, foreground/background transitions.
- **Trust relay:** implement relay modes: open relay, friends-only relay, and GPS-defined trusted zones.
- **Background RV/WCF:** keep rendezvous anchors and well-connected friends useful while the app is backgrounded.
- **OOB identity:** peer links for identity exchange, intent binding, and bootstrapping friendship/connection state.
- **GLP integration:** expose the stabilized transport and rendezvous behavior through the GLP Networking API.

Internet-scale UDX: possible extensions

Transport and discovery

- **BLE discovery optimization:** energy-aware scan/advertise duty cycling without losing too many encounters.
- **Dual-transport selection:** choose BLE or UDX per message based on reachability, latency, trust, cost, and payload size.
- **Cost-aware routing:** quantify bytes, battery, retries, relay load, and latency per delivered message.

Measurement

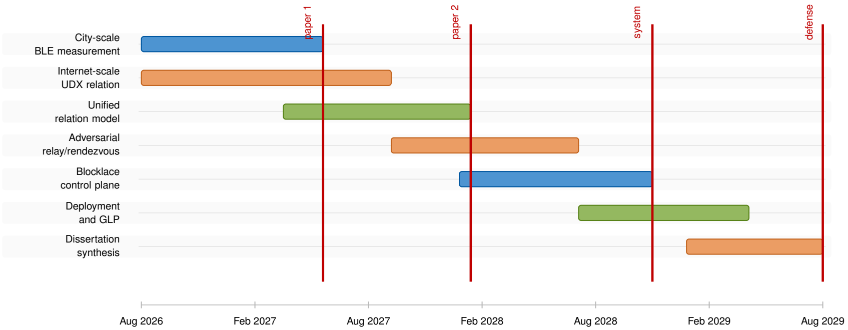
- **Well-connected friends:** characterize which peers can realistically help with signaling and rendezvous.
- **Grassroots NAT study:** measure NAT behavior, address churn, IPv4/IPv6 availability, and hole-punch success.

Trust and identity

- **Out-of-band identity exchange:** peer links should carry intent/attestation, not only a public key.
- **Friend-based relay:** safest relay mode; reduce attack surface with sealed forwarding and minimal visible metadata.
- **Geographic trust zones:** conferences, workplaces, or classrooms where local strangers may be acceptable relays.
- **Stranger relay:** highest-risk mode, but useful for emergency, protest, or disaster-connectivity scenarios.
- **Relay by trust level:** adapt forwarding policy to friend, trusted-zone, or stranger context.

Research question: once the system is stable, which relay and transport choices improve delivery enough to justify their cost and abuse surface?

PhD runway: two relation systems



- Thesis-to-PhD bridge: compare what grassroots relations mean under local physical mobility and under Internet-scale UDX connectivity.